Hierarchical POMDP: federated belief sharing across levels I

The flat sentinel world couples all agents at a single latent level — the creature's location. A natural extension is a **2-level hierarchical POMDP** in which location inference (Level 1, L1; 9 states) is coupled to a global *context* variable (Level 2, L2; 2 states: `quiet` / `alert`) that modulates the L1 prior. In the `alert` context the creature is expected near the den (center cell); in the `quiet` context the prior is uniform. Each sentinel runs alternating L1/L2 minimization (`fedference.pomdp.hierarchical_infer`) to infer both its location belief and the current context belief, then the colony federates both levels via a log-linear pool.

We compare two conditions over 960 seeded trials with 4 agents at sensor acuity 0.85:

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- **Flat** — agents ignore the hierarchy and infer location under a uniform prior;
- **Hierarchical** — agents run 4 alternating-minimization iterations to couple L1 and L2 beliefs before federating.

The measured location accuracies are 0.982 (flat) and 0.969 (hierarchical), a gap of -0.014. Across 128 independent seeds the hierarchical location accuracy is 0.974 (SD 0.005; 95 % CI 0.973–0.975) versus flat 0.982 (95 % CI 0.981–0.982), a mean accuracy gap of -0.008 (95 % CI -0.009—0.007; Wilcoxon signed-rank $p = 0.0000$, effect size $r = 0.940$, large). On location the gap is small but statistically reliable in the *negative* direction — the paired test rejects at $\alpha = 0.05$ and the gap's confidence interval (-0.009—0.007) excludes zero on the negative side — so the hierarchy does not improve location accuracy in this regime; if anything it pays a small, consistent location cost for carrying the extra latent level. Its added value is that it *also* infers the context latent, at accuracy 0.763 against a two-state chance baseline of 0.5. Two-level federation therefore runs L1/L2 inference end-to-end and resolves context above chance while paying a small, reliable location cost relative to the flat baseline (fig. 1). Context beliefs across the alternating-minimization iterations are shown in the top-middle panel: P(alert) sits above the two-state chance line and is stable from the first iteration onward when the observed location is the center cell — the center-cell observation pins the context posterior immediately, because the alert context-conditioned L1 prior is peaked there. The full construction and parameter sweep are detailed in the supplement (sec. 13.10). For the effect of acuity and colony size on these results, see sec. 13.13.

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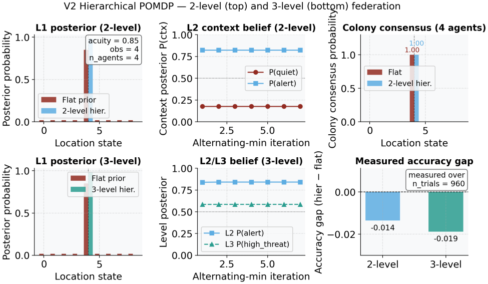

Figure 1: Six-panel hierarchical POMDP belief-dynamics and accuracy diagnostic. Source relation: source-inspired original project diagnostic for a hierarchical POMDP extension; estimand: posterior probabilities and final location-accuracy gap in the declared seeded protocol; uncertainty: deterministic seeded run, so no resampling interval is shown. Six-panel (2x3) visualization of the V2 hierarchical POMDP belief dynamics. Top row shows the 2-level world; bottom row shows the 3-level extension. Top-left panel: x-axis indexes the 9 location states; y-axis shows posterior probability for the flat-prior and 2-level hierarchical conditions given a single center-cell observation. Top-middle panel: x-axis is alternating-minimization iteration number; y-axis shows the L2 context posteriors P(quiet) and P(alert) under 2-level inference, pinned by the center-cell observation and stable from the first iteration onward. Top-right panel: x-axis indexes location states; y-axis shows the colony L1 consensus probability after federating 4 agents, comparing flat vs 2-level hierarchical. Bottom-left panel: x-axis indexes location states; y-axis shows posterior probability for the flat-prior and 3-level hierarchical conditions given a single center-cell observation. Bottom-middle panel: x-axis is alternating-minimization iteration number; y-axis shows L2 P(alert) and L3 P(high_threat) under 3-level inference, stable across iterations. Bottom-right panel: two bars showing the measured final location-accuracy gap (hierarchical minus flat) for the 2-level and 3-level systems, each a single scalar measured over 960 trials, with a zero reference line. Deterministic seeded run (seed 0), so bars carry no error band.